Bellman-Guided Warm-Start Search for LSTM Configuration in ECG Forecasting

Anonymous Author(s)
Affiliation withheld for blind review
Email: author@institution.edu

***Abstract*—Sequential hyperparameter searches often discard trained parameters between evaluations, even when consecutive candidates differ only locally. We examine whether conservative checkpoint inheritance can make a small Q-learning search budget productive for LSTM-based electrocardiogram (ECG) forecasting. Hidden width, recurrent depth, and dropout define an 18-state Markov decision process; seven coordinate-wise actions navigate this space. At each episode, the selected LSTM is trained for three epochs, evaluated on 5,000 validation windows, and assigned reward $r = -\mathrm{MSE}$. Parameters are inherited only when every checkpoint tensor is shape-compatible; otherwise, training restarts. The recorded CPU experiment uses 5,000 ECG5000 traces, 600,000 length-20 windows, and 12 episodes. Its best Adam-based warm-start checkpoint attains MSE 0.088472 and $R^2 = 0.9133$. Relative to the notebook's cold-start/manual-SGD run, MSE decreases from 0.943495 by $10.66\times$, whereas MAE decreases from 0.727927 to 0.195127 ($3.73\times$) at the same checkpoint. The tenfold factor therefore applies to MSE, not MAE. This comparison is not a causal estimate of the effect of RL: both runs use the same controller, while optimizer, inheritance policy, and effective training age change jointly. The empirical contribution is thus a transparent analysis of a warm-start search pipeline, together with the controls required to determine whether its apparent advantage generalizes.**

***Index Terms*—electrocardiogram, LSTM, Q-learning, Bellman equation, hyperparameter optimization, time-series forecasting, weight reuse**

I. Introduction

Electrocardiograms (ECGs) couple abrupt local morphology with longer temporal dependencies, making them a demanding setting for sequential prediction. Open resources such as PhysioNet [1] and the UCR Time Series Archive [2] support reproducible study of these signals. Although ECG5000 is typically used for classification, its traces also admit a self-supervised forecasting task: infer the next standardized amplitude from a fixed history. This task probes short-horizon temporal representation without treating the diagnostic class label as a target.

LSTM performance in this regime is governed not only by learned weights but also by architectural and optimization choices. Width controls state capacity, depth determines the recurrent hierarchy, and dropout regularizes inter-layer communication; optimizer and training duration alter the solution reached within any fixed architecture. Empirical comparisons of LSTM variants show that these apparently routine choices can dominate marginal changes to the cell design [3], [4]. A search procedure that retrains every nearby configuration from scratch may therefore spend most of its budget relearning transferable structure.

Existing HPO strategies address evaluation cost in different ways. Random search is a strong baseline [5]; Bayesian optimization uses a surrogate to prioritize informative evaluations [6]; successive halving allocates progressively larger budgets to promising candidates [7], [8]; and BOHB combines model-based proposal with bandit allocation [9]. RL-based NAS instead represents architecture design as a sequence of rewarded decisions [10]. These approaches motivate a narrower question: when candidate LSTMs differ by a single local edit, can a transparent controller exploit compatible weights without obscuring what is being compared?

We study this question in the finite design space encoded by the supplied notebook. A tabular Q-learning agent modifies one hyperparameter at a time, and a strict shape test determines whether the next candidate inherits the preceding checkpoint. The resulting method is deliberately modest in scale, but fully inspectable. Its contributions are:

- a finite-state formulation of LSTM width, depth, and dropout that exposes every controller transition;
- a conservative inheritance rule that reuses a checkpoint only under complete tensor-shape compatibility; and
- an evidence-calibrated analysis that separates the observed $10.66\times$ MSE reduction from the $3.73\times$ MAE reduction and from any claim of causal attribution to RL.

The experiment is an algorithmic case study rather than a clinical evaluation. Because the recorded comparison changes several factors simultaneously, we treat it as evidence about the end-to-end warm-start procedure, not as an ablation of Q-learning or weight inheritance in isolation.

II. Related Work

A. Deep Models for Time Series

LSTM introduced gated memory as a mechanism for preserving gradients over long sequences [3]. Later empirical work showed that robust baselines and tuned components often matter more than increasingly elaborate gate variants [4]. Hybrid architectures such as LSTM–FCN combine recurrent and convolutional pathways [11], while large-scale evaluations across UCR datasets document substantial sensitivity to architecture and preprocessing [12]. Forecasting surveys likewise conclude that recurrent models remain competitive when validation and model selection are handled carefully [13], [14].

Our forecaster is intentionally minimal: an LSTM maps a univariate window to its final hidden state, and a linear head predicts the subsequent sample. Dropout is searchable because it can limit co-adaptation [15]; under the recurrent implementation used here, however, it acts only between stacked layers and is inactive for $L = 1$.

B. Sequential Configuration Search

Bellman's dynamic-programming view of sequential decisions [16] underlies Q-learning, which estimates action values without an explicit transition model [17]. RL-based NAS applies this principle to architecture generation [10]. ENAS reduces its cost through shared parameters [18], while network-transformation methods preserve compatible weights as a graph evolves [19]. Such reuse is computationally attractive, but it couples a candidate's measured quality to its optimization history.

The present search occupies a deliberately restricted point between black-box HPO and full NAS. Each action edits at most one coordinate of a fixed LSTM family, so the transition graph is explicit and the value function remains tabular. Checkpoint reuse is sequential rather than supernet-based and is permitted only when all parameter shapes agree; Adam [20] then performs the short local update. This restriction sacrifices architectural breadth, but makes the interaction between controller state, model compatibility, and accumulated training directly auditable.

III. Methodology

A. Problem Statement

Let $\mathcal{D}_{\mathrm{tr}}$ and $\mathcal{D}_{\mathrm{val}}$ denote the training and validation sets of input windows and next-sample targets. For an LSTM configuration $\lambda = (H, L, D)$, training procedure $\mathcal{A}$, and parameters θ, the validation objective is

$$\mathcal{L}_{\mathrm{val}}(\lambda, \theta) = \frac{1}{|\mathcal{D}_{\mathrm{val}}|} \sum_{(\mathbf{X}, y) \in \mathcal{D}_{\mathrm{val}}} \left(y - f_{\lambda,\theta}(\mathbf{X})\right)^2. \quad (1)$$

Conventional HPO seeks $\lambda^* = \arg\min_{\lambda \in \Lambda} \mathcal{L}_{\mathrm{val}}(\lambda, \mathcal{A}(\lambda))$ under an implicit independence assumption: each λ is trained without reference to previous candidates. Checkpoint inheritance violates this assumption. The observed loss becomes path-dependent because it reflects both the current architecture and the optimization history carried into it. We therefore represent search as a finite-horizon decision process with return

$$G_k = \sum_{j=0}^{K-k-1} \gamma^j r_{k+j+1}, \qquad r_{k+1} = -\mathcal{L}_{\mathrm{val},k+1}, \quad (2)$$

over $K = 12$ episodes. The reported model is the checkpoint with minimum recorded validation MSE rather than the terminal state; this decouples deployment from a potentially inferior final exploratory action.

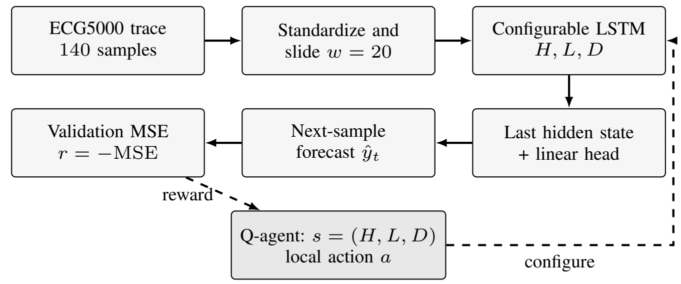

Fig. 1. Block-wise architecture of the Q-guided ECG forecaster. Solid arrows form the prediction path; dashed arrows close the search loop.

B. Forecasting Task and LSTM

Let $\mathbf{x} = (x_1, \ldots, x_{140})$ be a standardized ECG trace. With window length $w = 20$, each valid position yields the supervised pair

$$\mathbf{X}_t = [x_{t-w}, \ldots, x_{t-1}], \qquad y_t = x_t. \quad (3)$$

The LSTM consumes $\mathbf{X}_t$ sequentially. Its gated update can be written compactly as

$$\begin{aligned} \mathbf{i}_t &= \sigma(W_i x_t + U_i \mathbf{h}_{t-1} + \mathbf{b}_i), \\ \mathbf{f}_t &= \sigma(W_f x_t + U_f \mathbf{h}_{t-1} + \mathbf{b}_f), \\ \mathbf{o}_t &= \sigma(W_o x_t + U_o \mathbf{h}_{t-1} + \mathbf{b}_o), \\ \tilde{\mathbf{c}}_t &= \tanh(W_c x_t + U_c \mathbf{h}_{t-1} + \mathbf{b}_c), \\ \mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \end{aligned} \quad (4)$$

The final hidden state is mapped to $\hat{y}_t = \mathbf{w}^\top \mathbf{h}_t + b$ by a linear regression head. Candidate quality is measured by

$$\mathrm{MSE} = \frac{1}{N} \sum_{n=1}^{N} (y_n - \hat{y}_n)^2, \quad \mathrm{MAE} = \frac{1}{N} \sum_{n=1}^{N} |y_n - \hat{y}_n|. \quad (5)$$

C. LSTM Configuration as an MDP

The controller state $s = (H, L, D)$ records the hidden width, number of recurrent layers, and dropout rate:

$$H \in \{32, 64, 128\}, \quad L \in \{1, 2\}, \quad D \in \{0, 0.2, 0.4\}. \quad (6)$$

The Cartesian product contains 18 states. Seven actions increase or decrease a single coordinate, or preserve the current configuration. Boundary clipping maps infeasible outward moves to self-transitions. Figure 1 summarizes the resulting closed loop: the deterministic transition $T(s, a)$ configures the forecaster, whose validation loss supplies the controller reward.

After the LSTM specified by $s' = T(s, a)$ is trained, the environment returns $r = -\mathrm{MSE}_{\mathrm{val}}$. The agent then applies the one-step Q-learning update

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right], \quad (7)$$

with learning rate $\alpha = 0.4$ and discount factor $\gamma = 0.8$. An ϵ-greedy policy ($\epsilon = 0.5$) alternates uniformly random actions with greedy selection. All action values are initialized to zero; the trajectory begins at $(64, 1, 0)$ and terminates after 12 episodes.

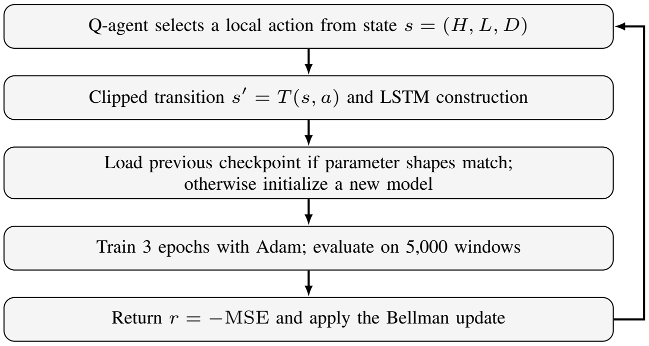

Fig. 2. One episode of the warm-start Q-learning search.

Algorithm 1 Warm-Start Q-Learning for LSTM Configuration

Require: $\mathcal{D}_{\text{tr}}$, $\mathcal{D}_{\text{val}}$, state set $\mathcal{S}$, actions $\mathcal{U}$
Ensure: Best checkpoint b^* and episode log $\mathcal{B}$

```
Set s_0 = (64, 1, 0), Q(s, a) = 0, θ⁻ ← ∅, B ← ∅
for k = 1, ..., 12 do
    Select a_k by an ε-greedy policy, ε = 0.5
    Set s_{k+1} ← T(s_k, a_k) and construct its LSTM
    Load θ⁻ if parameter shapes match; otherwise reinitialize
    Train three epochs with Adam on D_tr
    Evaluate MSE, RMSE, MAE, and R² on D_val
    Set r_k ← −MSE_k and update Q using (7)
    Save the checkpoint; append (s_{k+1}, a_k, r_k) and metrics to B
end for
return b* ← arg min_{b∈B} MSE(b)
```

D. Compatible-Weight Reuse

Figure 2 details a search episode. After constructing the candidate specified by s', the procedure compares every state-dictionary tensor with the preceding checkpoint. Complete shape agreement permits exact loading followed by three additional epochs; any mismatch triggers full reinitialization. Repeated visits to $(128, 1, 0)$ in episodes 3–6 therefore form one continued optimization trajectory. A dropout change at $L = 1$ also passes the shape test, although it has no operational effect because recurrent dropout is inactive. In contrast to ENAS-style supernet sharing [18], reuse is local in time and no parameters are shared concurrently across candidates.

IV. Experimental Protocol

The recorded pipeline downloads the ECG5000 train and test files from the UCR archive, concatenates all 5,000 traces, and removes the class labels from the 140-sample signals. Each temporal position is standardized over the resulting matrix. Equation (3) then yields 120 examples per trace and 600,000 windows overall. The notebook applies a seeded 80/20 split at the window level and, for CPU feasibility, retains the first 20,000 training and 5,000 validation windows. Table I reports the remaining settings. Section VI discusses the inferential consequences of this preprocessing protocol.

Both recorded searches share the state space, action set, and reward. The comparison run reinitializes every candidate and performs three epochs using a manually implemented SGD update; the warm-start run jointly introduces Adam and compatible checkpoint inheritance. MSE, root MSE (RMSE), MAE, and R^2 are evaluated on the validation subset. Because the notebook omits elapsed time, initialization and action-selection seeds, repeated trials, confidence intervals, and an untouched final test result, our analysis is restricted to the recorded trajectory.

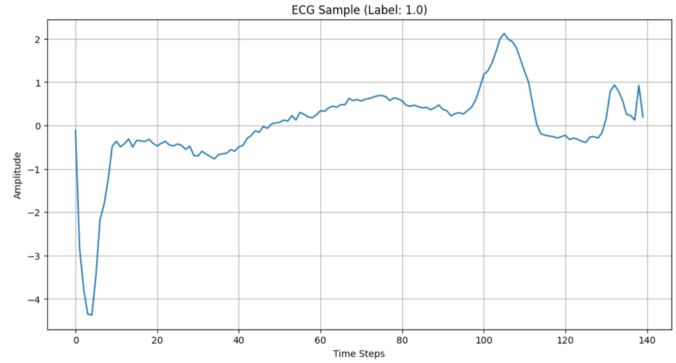

Fig. 3. An ECG5000 trace visualized in the source notebook. The class label is not used by the forecasting target.

TABLE I
Recorded Notebook Configuration

Setting	Value
Dataset / traces / length	ECG5000 / 5,000 / 140
Forecast window / target	20 / next sample
Generated / used windows	600,000 / 20,000 train, 5,000 val.
Batch size / device	512 / CPU
Episodes / epochs per episode	12 / 3
Agent $(\alpha, \gamma, \epsilon)$	$(0.4, 0.8, 0.5)$
Warm-start optimizer / rate	Adam / 10^{-3}
Baseline optimizer / rate	manual SGD / 10^{-3}
Selection objective	minimum validation MSE

In addition to (5), the recorded metrics are

$$\text{RMSE} = \sqrt{\text{MSE}}, \qquad R^2 = 1 - \frac{\sum_n (y_n - \hat{y}_n)^2}{\sum_n (y_n - \bar{y})^2}. \tag{8}$$

MSE serves as both the training-aligned selection objective and the controller reward. MAE provides a complementary linear-error measure, while RMSE restores the scale of the standardized target. Because RMSE is the square root of MSE, a $10.66\times$ MSE reduction necessarily corresponds to a $\sqrt{10.66} = 3.27\times$ RMSE reduction rather than another tenfold change.

A. Model Size and Evaluation Budget

For scalar input and output, a one-layer LSTM contains $4H^2 + 13H + 1$ trainable parameters. A two-layer model has $12H^2 + 21H + 1$, including both recurrent layers and the linear head. As Table II shows, the design space ranges from 4,513 to 199,297 parameters. Width and depth therefore change the memory and computational requirements substantially; dropout changes neither the parameter count nor the tensor shapes.

With 20,000 training windows and a batch size of 512, each epoch contains 40 mini-batches, including one partial batch.

TABLE II
TRAINABLE PARAMETERS BY WIDTH AND DEPTH

H	$L=1$	$L=2$	Two-/one-layer ratio
32	4,513	12,961	2.87×
64	17,217	50,497	2.93×
128	67,201	199,297	2.97×

TABLE III
SELECTED WARM-START SEARCH EPISODES

Ep.	(H, L, D)	MSE	RMSE	MAE	R^2	Reuse
1	$(32, 1, 0)$	.2031	.4506	.2720	.8011	No
3	$(128, 1, 0)$	.1350	.3674	.2471	.8678	No
4	$(128, 1, 0)$	.0996	.3156	.2021	.9024	Yes
6	$(128, 1, 0)$	**.0885**	**.2974**	.1951	**.9133**	Yes
8	$(64, 2, 0)$	.1281	.3579	.2267	.8746	No
12	$(64, 1, .2)$	.0910	.3017	**.1932**	.9109	Yes

An episode therefore performs 120 gradient updates, and the full search performs 1,440 updates across all instantiated models. When recurrent matrix multiplication dominates, the cost of a length-w window is approximately $O(wLH^2)$. A larger study should consequently report elapsed time, energy, parameter count, and update budget alongside forecast error. The present reward optimizes accuracy alone and does not prefer a smaller model when two candidates have equal loss.

V. RESULTS AND DISCUSSION

A. Search Trajectory

Figure 4 reveals a trajectory governed as much by parameter continuity as by architectural choice. Episode 1 selects $(32, 1, 0)$ and attains MSE 0.203070; subsequent actions expand the hidden state to 64 and then 128 units. Once $(128, 1, 0)$ is retained, episodes 3–6 inherit the same checkpoint. Over these four evaluations, MSE decreases monotonically from 0.134992 to 0.088472, while R^2 rises from 0.8678 to 0.9133. Structural changes in episodes 7 and 8 break compatibility, reset optimization, and degrade all error metrics. The selected checkpoints in Table III are therefore best interpreted as states along a coupled architecture–training trajectory, not independent draws from the design space.

The minimum-MSE checkpoint contains 67,201 parameters, or 33.7% of the 199,297-parameter maximum. The search consequently does not collapse to the largest available network. Episode 8 provides a useful contrast: a two-layer, 64-unit LSTM with 50,497 parameters yields MSE 0.128084, substantially worse than the continued one-layer, 128-unit trajectory. This difference is descriptively informative but not an architecture ablation, because the models enter evaluation with different parameter histories and cumulative update counts.

B. Magnitude and Meaning of the Improvement

Table IV compares the checkpoints selected by minimum validation MSE. The cold-start/manual-SGD run selects $(128, 1, 0.2)$; because $L = 1$, its nominal dropout is inactive. The warm-start/Adam run selects $(128, 1, 0)$ at episode 6. MSE decreases from 0.943495 to 0.088472, equivalent to

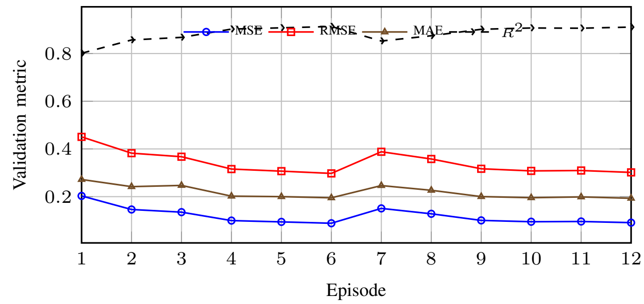

Fig. 4. Recorded validation metrics across the 12 warm-start episodes.

TABLE IV
MSE-SELECTED CHECKPOINT COMPARISON

Search variant	MSE	RMSE	MAE	R^2
Cold start + manual SGD	.943495	.971337	.727927	.0759
Warm start + Adam	**.088472**	**.297443**	**.195127**	**.9133**
Reduction	90.62%	69.38%	73.19%	+0.8374
Factor	10.66×	3.27×	3.73×	—

a 90.62% reduction or a factor of 10.66×. At the same checkpoint, RMSE falls by 69.38% (3.27×) and MAE by 73.19% (3.73×). Optimizing each metric separately changes the selected episode: the minimum MAE, 0.193173, occurs at episode 12 and represents a 3.77× reduction.

The factor must be stated with its metric: the evidence supports a tenfold MSE reduction, not a tenfold MAE reduction. More importantly, Table IV is an observational contrast between two pipelines rather than an estimate of a single treatment effect. Q-learning is common to both. The warm-start run simultaneously changes the optimizer, initialization policy, and effective training age: $(128, 1, 0)$ receives 12 cumulative epochs across episodes 3–6, whereas a cold-start candidate receives three. No counterfactual in the recorded experiment can separate these contributions, so attribution is limited to the joint procedure.

The increase in R^2 from 0.0759 to 0.9133 is consistent with the same conclusion: predictions from the selected warm-start checkpoint track the variance of the recorded validation windows much more closely. Yet the windows are standardized and may share overlapping temporal context within a trace. The statistic quantifies fit to this validation construction; it does not establish patient-level generalization, morphology preservation, or diagnostic utility.

C. Practical Interpretation

Sequential reuse changes the object being optimized. Under cold starts, a state's reward approximates the quality of an architecture after a fixed budget; under warm starts, it reflects architecture, inherited representation, and accumulated compute. The latter may be more efficient operationally, but it is no longer a stationary architecture score. A principled extension should augment the state with checkpoint age or consumed updates and impose a common compute budget at comparison time.

The controller also exhibits implicit optimistic exploration. Because rewards are negative and Q is initialized to zero, an unvisited action can dominate an observed action even in the greedy branch. This bias may be useful for coverage, but a 12-episode trajectory cannot explore the 18×7 state–action space or justify convergence. Randomized tie breaking, a decaying ϵ, explicit visitation logs, and separate search and selection data are necessary for a statistically interpretable study.

Finally, tensor-shape agreement is a necessary but incomplete notion of transferability. A dropout change at $L = 1$ preserves every tensor yet changes no active recurrent operation, whereas a depth change invalidates the entire checkpoint despite substantial shared structure. A richer operator could copy common submodules, initialize only newly introduced components, and expose the inherited parameter fraction as part of the search record, following network-transformation methods [19].

VI. Threats to Validity and Future Work

The dominant internal-validity threat is information coupling between training and validation. Standardization precedes partitioning, and overlapping windows from the same trace are assigned independently to both pools. Consequently, the scaler contains validation information and near-duplicate temporal contexts may cross the split. A defensible protocol must partition complete traces first, fit preprocessing exclusively on the training partition, preserve the official test set, and evaluate the selected model once on untouched data.

Statistical evidence is limited to one 12-episode trajectory over truncated subsets. Repeated searches with declared seeds, paired trace-level partitions, and uncertainty intervals are required to establish stability. Causal attribution further requires a factorial design that independently varies optimizer, inheritance, and training budget, together with equal-budget random search, Hyperband, BOHB, and a fixed-configuration LSTM. Table V summarizes these controls.

External validity is narrower still. Next-sample error on standardized ECG5000 traces is neither a diagnostic endpoint nor a measure of clinical benefit. Translation would require patient-level cohorts, morphology-aware outcomes, external validation, and explicit accounting of inference cost. Search performance and final generalization should be reported separately: the former on validation data used by the controller, the latter once on a sequestered test set.

A. Reproducibility and Ethical Scope

The notebook records the data sources, split seed, subset sizes, implementation, controller constants, and per-episode metrics. Exact replay additionally requires pinned software versions, complete Python/NumPy/PyTorch seeds, the realized action sequence, all intermediate checkpoints, hardware metadata, and a machine-readable results file. These artifacts are also prerequisites for comparing computational cost rather than validation loss alone.

TABLE V
Validity Risks and Required Remedies

Risk	Likely effect	Required remedy
Window-level split	correlated train/validation contexts	split by trace before scaling/windowing
Full-data scaling	validation information in normalization	fit scaler on training traces only
One stochastic run	unknown variance and selection stability	multiple seeds and confidence intervals
Unequal warm-start age	confounds architecture and training budget	compare equal updates or add age to state
Optimizer changed	gain cannot be assigned to reuse or RL	factorial Adam/SGD and reuse ablation
Validation-only report	optimistic post-selection estimate	untouched final test and external cohort

The system produces neither a diagnosis nor a patient-specific recommendation. ECG5000 class labels are excluded and the target is a single standardized amplitude. Forecast error therefore has no direct interpretation as sensitivity, specificity, calibration, or safety. The method should be regarded strictly as a signal-modeling study unless evaluated against clinically defined endpoints under appropriate governance and expert review.

VII. Conclusion

We formulated LSTM configuration as a finite Bellman search in which compatible checkpoints persist across local architectural edits. The recorded 12-episode ECG5000 trajectory reaches MSE 0.088472 and $R^2 = 0.9133$; relative to the accompanying cold-start/manual-SGD run, MSE and MAE improve by $10.66\times$ and $3.73\times$, respectively, at the selected checkpoint. The central observation is that checkpoint inheritance can convert repeated configurations into continued optimization. Its benefit, however, is inseparable here from Adam and unequal cumulative training. The result should therefore motivate a controlled study, not a causal claim. Trace-level partitioning, repeated paired searches, factorial equal-budget ablations, and one sequestered final test are the necessary next steps.

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